

INTERNSHIP REPORT



SPARSH Super Speciality Hospital
146, Infantry Rd, Vasanth Nagar,
Bengaluru, Karnataka 560001

SUBMITTED BY:

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BSC Nutrition and dietetics

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Bengaluru – 560027

Duration: 10th Feb 2026 – 23 Feb 2026

SPARSH SUPER SPECIALITY HOSPITAL

#146, Infantry Road, Bangalore-560001

CERTIFICATE

This is to certify that "**HOSPITAL INTERNSHIP REPORT**" is a Bonafide Hospital case Study Work carried out by

Ms. Shaik Hazira Tamreen under the supervision of Chief Dietitian Mrs. Sheela Joseph and Junior Dietitian Ms. Kaveri, Sparsh Super Specialty Hospital, Bangalore, in partial fulfilment of 6th Semester in BSc. Nutrition and Dietetics under Bishop Cottons Women's Christian College during the period of 10th February 2026 to 23rd February 2026.

Date: 23/02/2026

Signatures:

Chief Dietitian:

1) Mrs. Sheela Joseph

Junior Dietitian:

2) Ms. Kaveri

Junior Dietitian:

3) Ms. Rakshitha

ACKNOWLEDGEMENT

First and foremost, we are grateful to God for bestowing upon us the strength that enabled us to successfully finish the internship course.

We would like to express our heartfelt appreciation to the administration of Sparsh Super Specialty Hospital Infantry Road Bangalore for providing us with the opportunity to intern at this prestigious institution.

We would like to thank the Chief Dietitian, Mrs. Sheela Joseph for her excellent guidance and suggestions during an internship which would not have been possible without her assistance. We would also like to thank Ms. Kaveri for her assistance.

We would like to thank the staff of the food service and beverage department for their cooperation and effort

We owe our immense gratitude to Dr Marie Kavitha Jayakaran, Head of the Home Science Department of Bishop Cottons Women's Christian College, Autonomous, for her support and encouragement.

Finally, we would like to express our gratitude to our family and friends for their assistance, inspiration, and unwavering support. All of which enabled us to successfully complete the hospital internship.

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INTRODUCTION

SPARSH Hospital is one of the best multispecialty hospitals in Bangalore. It has various branches all over Bangalore: Hosur Road, Yeshwanthpur, RR Nagar, Infantry Road and is spread across Hassan and Davangere.



Its first branch was established in 2006 at Bommasandra and its super-specialty branch at Yeshwanthpur (SPARSH Super Specialty Hospital) was established in 2015. With the aim of extending its services across specialties, SPARSH Super Specialty Hospital was launched in May 2015 in Yeshwantpur. Within a year, it has exceeded expectations by successfully completing over 1000 surgeries and treating over 10,000 outpatients. The hospital is now considered a one-stop destination for all diagnostic and treatment needs.

This hospital is now reputed as a destination for Orthopedic surgery, knee replacement surgery. sports medicine. aesthetic surgery and much more. The team has expertise in trauma care, joint replacement, sports injuries, limb deformity corrections arthroscopy. The team is supported by appropriate interventional radiology and the best vascular, plastic surgical and neurosurgical specialists under one roof.

The hospital is considered as a one-stop destination all treatment and diagnostic needs.

Apart from Blood Bank, Laboratory Services and Health Checks It has 24/7 accident and emergency services. Sparsh Super Specialty Hospital has a separate wing for OBG and Pediatrics, with Birthing Suites, Labor OT and Labor Rooms. SPARSH is committed to constantly explore opportunities and develop processes and systems by which healthcare could become affordable in India.



SPARSH Hospital for Women and Children was envisioned with the thought that women and children should be able to access world class health care under one roof without compromising exclusivity. Out of this thought was born the idea of having a separate unit for women and children but still being able to access all the other specialties of medicine if required.

This dedicated multispecialty unit exclusively for women and children offers special and personalized care covering all age groups, from neonates to geriatrics. While today women have more freedom to create, push beyond their boundaries and make an impact. Their health care too, needs a new level of service which is all-inclusive, dynamic and non-judgmental.

Here, they provide solution-driven care through their diverse talent of experts in a collaborative environment. When faced with any complexity. They have both the infrastructure and the expertise to resolve emergencies round the clock.

SPARSH Hospital does not only treat patients within the country but also treats patients from across the globe. SPARSH Hospital is a NABH accredited hospital and is one of the most respected hospital groups in the country. They have been instrumental in revolutionizing the healthcare sector in India and continue to do so till date.

The Sparsh Hospital "touch of life is an experience that is imprinted in the minds of all those who they have cared for in the last two decades. Sparsh hospital is about commitment and adherence to the minutest of details. For them, there is nothing more important than upholding the principles of equality, fairness and non-discrimination in treating their patients.

New opportunities, challenges and the drive to help more people is what pushes them to be the best. Ever since the inception, they have had one goal: to make quality healthcare accessible to the poor. At SPARSH, they stand for achieving clinical excellence with social relevance. Making a difference and positive impact on society at a larger scale has always been vital to them.

From the beginning, SPARSH has achieved many milestones that have led them in taking pride in their reputation. They have performed complex surgeries and pushed themselves every day, aiming to live up to the endeavor of touching as many lives as possible. Within a year of their foundation, they have performed over 1000 surgeries and became a name associated with quality healthcare.

The pillars of SPARSH HOSPITAL's path breaking approach include:

- Practicing Evidence-based medicine with the highest integrity.
- Educating patients and their families for participative decision making.
- Focusing on Enhancing skill sets through learning and training.
- Harnessing the advances in science and technology to deliver quality health care.

VISION:

To be a leader in health care and serve as a destination for complexities.

MISSION:

- Practice evidence-based medicine with highest integrity.
- Educating patients and their families for participative decision making.
- Focus on enhancing skill sets through learning and training.
- Harness the advances in science and technology to deliver quality health care.

ABOUT THE CHAIRMAN

Dr. Sharan Shivraj Patil
Chairman and Chief Orthopedic Surgeon
MBBS. M. S. (Orth), MCH (Orth), Liverpool,
FRCS (England)

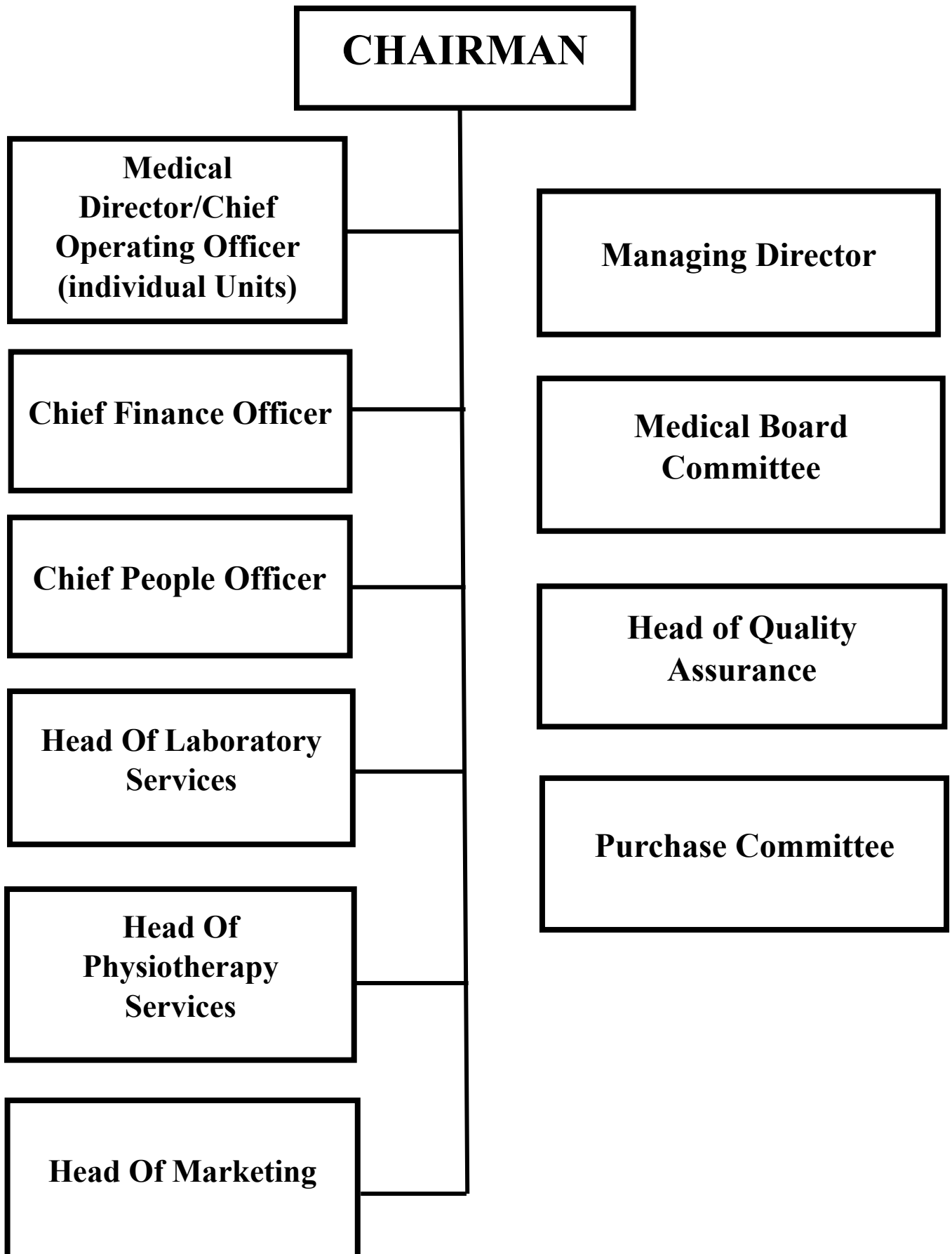
Specialities: Orthopedic



Dr. Sharan completed his medical school with an academic distinction from MR Medical College, Gulbarga after a brief stint at St Martha's Hospital in Bangalore. His post-graduate education was at Kasturba Medical College Manipal. He was awarded D'Othro in 1990 followed by the MS (Ortho) in 1991 with a gold medal. In 1992, Sharan moved to the UK for further training. In 1995, Sharan was bestowed with the coveted degree of MCH. Ortho from Liverpool University, making him the youngest graduate to be accorded this honor since the course began in 1926. Sharan returned to India in 1996 and joined the premier Institute of Manipal Hospital in Bangalore as their consultant in Orthopedics. Whilst working with this organization, he single-handedly performed over 5000 major complex surgical procedures, especially in the field of Pediatric Orthopedics, Joint Replacement and Complex Trauma. This earned him the impeccable reputation of delivering high-quality orthopedic care.

To pursue the dream of taking world standard medical care to the masses, SPARSH Hospital was established in the year 2006 with 120 beds and 5 operating rooms with a capacity of carrying out over 30 surgeries a day. Dr Sharan has been a pillar in the growth of SPARSH. With the aim of providing Tier 1 quality health care and making it available and affordable, he has led SPARSHA to expand to different units

ORGANIZATION STRUCTURE- CORPORATE ORGANOGRAM



HOSPITALS

```
graph TD; A[HOSPITALS] --> B[HEALTH CITY  
Sparsh Hospital  
accidents, orthopedics, plastic surgery and maxillofacial 1 surgery]; A --> C[INFANTRY ROAD  
Sparsh Hospital for advanced surgeries]; A --> D[HASSAN  
Sparsh Hospital]; A --> E[RR NAGAR  
Sparsh Hospital]; A --> F[YESHW-ANTPUR  
Sparsh Super Speciality Hospital]; A --> G[DAVAN-GERE  
Sparsh hospital];
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HEALTH CITY

Sparsh Hospital
accidents, orthopedics, plastic surgery and maxillofacial 1 surgery

INFANTRY ROAD

Sparsh Hospital for advanced surgeries

HASSAN

Sparsh Hospital

RR NAGAR

Sparsh Hospital

YESHW-ANTPUR

Sparsh Super Speciality Hospital

DAVAN-GERE

Sparsh hospital

WORKING OF THE DIETARY DEPARTMENT

INTRODUCTION:

Dietetics is an application of science and art human nutrition in helping people to select and obtain food for the primary purpose of nourishing their bodies in health and disease throughout the life cycle. Clinical dietetics mainly deals with using food as an agent in effective recovery from disease.

The department of dietetics is a paramedical department which forms an integral part of inpatient therapeutic care during their hospital stay.

PURPOSE:

- To plan a nutritionally well-balanced diet for all in-patients of the hospital. keeping in mind their medical conditions. Modifications are made according to their palate and religious belief.
- To educate all inpatients about specific diet during hospitalization and at the time of discharge. All inpatients on therapeutic diets are given customized diet sheets to follow at home.

SCOPE OF SERVICE:

- Inpatient food services
- Cafeteria
- Staff Food Service

QUALITY OBJECTIVE:

- To provide nutritious and hygiene food to inpatient/outpatients and visitors/staff of the hospital.
- Meals service to inpatients is the highest priority of the food service department, each inpatient is provided with the meal and refreshments that are to be prescribed and approved by the attending dietitian.
- The serving time of providing a dietary meal or refreshment is to be closely monitored. • The kitchen/service staff, IPD team and dietitians work in close coordination to achieve the indicated serving times.

FUNCTIONING:

- Dietitians Work in two shifts 1) 7am-3pm 2) 11 am-7 pm
- Dietitians in the morning shift taste and do sensory evaluation on all breakfast items which are served to the patients and record their comments in Food Assessment Record where every major meal is evaluated. She also checks tube feed preparation.

- The dietitian formulates diets for new in-patients after collecting diet request slip.

SPARSH HOSPITAL
THE TOUCH OF LIFE
146, Infantry Road, Bangalore - 560 001.

Name of the Patient : _____
UHID : _____

Name of the Consultant : Dr. Divind Bhatia
Diagnosis : L4-L5
Diet : Normal Diet

New Admission / Change of Diet / Bd Transfer : 105A

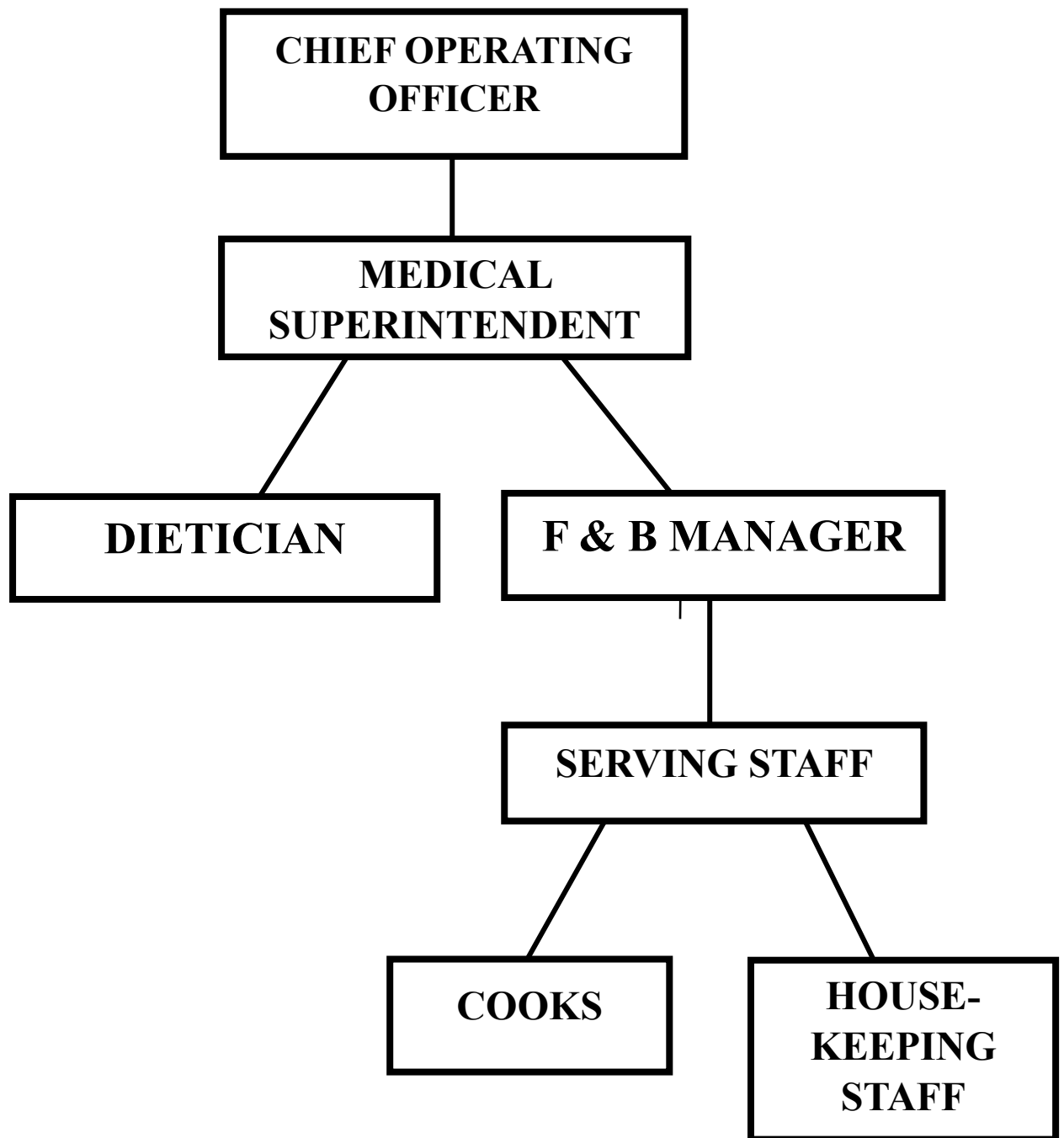
Nurse's Name : [Signature]
Time : 12 pm

Receiver's Name : _____
Time : _____

10903084
B/F No. 08/24/2413
Date of Admission: 2 Jun 2025 | 11:05 AM
Practitioner: Dr. DIVIND BHATEJA
Specialty: NEUROSURGERY & SPINE SURGERY
Ward & Bed No.: LEFTWING FIRST FLOOR RT | 105-A
Payor: Self Pay

- This is followed by ward rounds, files are checked for clinical findings, lab investigations and diet orders given by the doctors. All these are recorded in the inpatient diet list.
- All new admissions areas are assessed nutritionally at the time of admission and information is recorded in the initial nutritional assessment sheet. Patients asked if the food provided earlier was acceptable or not. Any changes required by the patients in the diet are noted down.
- Cards of each patient are prepared with their daily diet details, which are handed over to kitchen supervisor for service. Information provided by patients concerning changes in diet is also included in the cards. The steward dishes out the food for the patients into the trays under the supervision of the kitchen supervisor and dietitian and the food is distributed to the patients.
- The supervisor ensures all the patients have received their correct diet on time and notes down any new admission which will be informed in the dietitian for her record.

ORGANOGRAM- DIETARY ASSESSMENT



ROUTINE HOSPITAL DIET

- **Clear fluid:** This provides calories and electrolytes to prevent dehydration.
Example: Coconut water, Barley water, Arrowroot Kanji, Clear soup and Strained fruit juices.
- **Fully liquid:** This provides oral nutrition that is well tolerated by patients who cannot swallow or chew.
Example: Water, Fruit juices, Soup water and Strained vegetable soups.
- **Semi-solid diet:** This is a transition between liquid and solid diet.
Example: Porridge, Thin khichdi, Mixed rice with Rasam and Curd.
- **Soft-solid diet:** This diet has a soft texture.
Example: Well cooked, Soft and Steamed food like still, dal, Double boiled rice and Soft-boiled rice.
- **Normal Diet:** This is a balanced diet that contains optimal amounts of foods from each group to meet the patient's nutritional needs.
- **Weight Loss/Calorie Restricted Diet:** Calorie intake on this diet is restricted to a specific level. The amount of fat and sugar are limited.
- **Diabetic diet:** This is a diet that is properly distributed throughout the day and in such amounts that it helps achieve optimal caloric needs and maintain normal blood sugar levels.
- **Low fat, low cholesterol diet:** This is a diet in which the amount is limited to specific levels and a type of fat containing less saturated bonds is recommended. These diets are designed to lower elevated levels of serum cholesterol and other lipids to reduce cardiac disease.
- **Salt-free and sensory diet:** Indicated for people with renal, hepatic or cardiac disease, where the use of salt is limited or completely salt-free.
- **Neutropenic diet:** This is a sterile diet that provides only well-cooked foods and excludes foods that are potentially pathogenic. It is recommended for patients who are post-transplant or undergoing chemotherapy.
Example: Cottage cheese, Buttermilk, Fresh fruit, Fruit juices and Vegetable salads.
- **Renal Diet:** A standard diet modified for protein, sodium, potassium, and fluid to meet the patient's needs.
- **Ryle tube feeding:** Liquid or blended diets are designed to provide essential nutrients in a form that easily passes through the tube. It is administered through a nasogastric tube or percutaneous endoscopic gastrostomy (PEG) or percutaneous endoscopic jejunostomy (PEJ). They are formulated to meet specific therapeutic needs.
- **Cardiac diets:** Complex Carbohydrates, rich in liberal, salt-restricted, low-fat, low-cholesterol diets. Add MUFA. PUFA and fluid restriction.
- **Pier in pregnancy and breastfeeding:** High-calorie, protein-rich and nutritious foods. There are no restrictions. All foods can be included unless the patient has gestational diabetes or other medical complications.

Sample Diet Slip:



DIET CARD

UHID No.

Patient Name :

Age :

Diet : Normal / Diabetic / Low Salt Diet / Others :

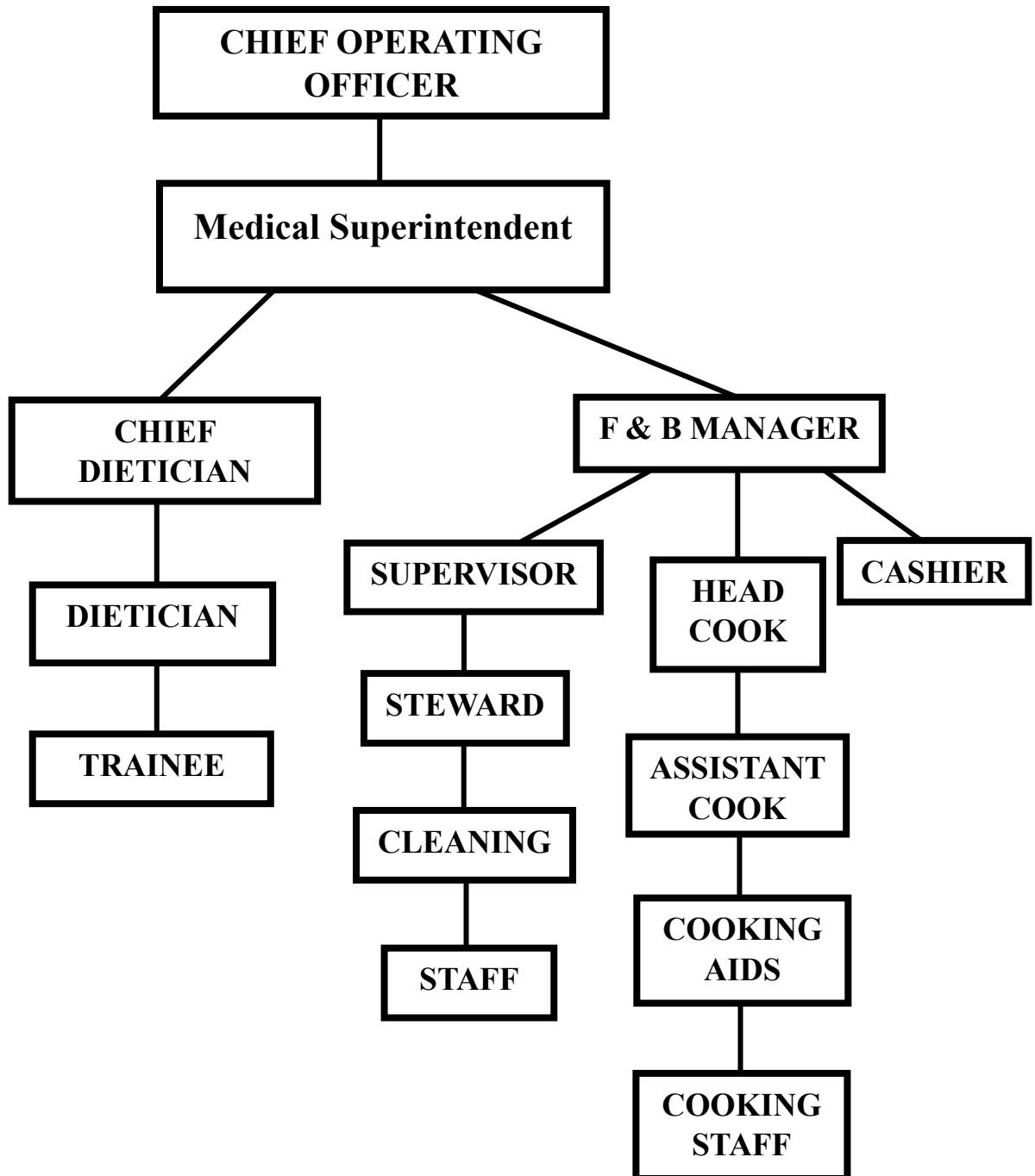
Early Morning	Break-fast	Post B/f	Mid Morning	Lunch	Tea	Dinner	Bed Time
Coffee	Cereal	Milk	FJ	DB Rice	Tea	DB Rice	Milk
	Fruit	FJ	BM	Chapati	Coffee	Chapati	BM
	Boiled Egg	TCW	Salad	Ragi Ball/	Milk	Ragi Ball/	
Tea	Toast			Porridge	Biscuits	Porridge	
Milk	Butter			Curd Rice		Broken-Wheat	
FJ	Cheese			Dhal		Kanji	
	Jam / Egg			Veg. Dry		Broken-Wheat	
TCW				Veg. Gravy		Khichidi	
	Cornflakes			Curd		Curd Rice	
	Oats			Fruit		Khichidi	
	Porridge			Broken		Sambhar	
				Wheat		Rasam	
						Veg.	
						Salad	
						Fruit	

Hospital Menu

Service	Breakfast	Mid-Morning	Lunch	Dinner
Days	7:30-8:30AM	10:30-11:00AM	1:00PM-1:30PM	7PM-8PM
Sunday	Cut fruits/idly/sambhar/Onion Tomato chutney	Soup Cream corn or Juice (carrot & beetroot)	Dal maharani/ malai paneer/ snake gourd poriyal/ chapati/ double boiled rice/ plain curd	Carrot kosambari/ radish sambhar/ melagu rasam/ DB rice/ yellow dal tadka/ tindly & peanut palya/ wheat dosa
Monday	Cut-fruits/Sprouts dosa/sambhar/coriander chutney	Soup palak shorba / juice (muskmelon)	Nei parupu/ diwani/ soya gobi adaraki/ chapati/ double boiled rice/ plain curd	Mixed salad/ soppu sambhar/ garlic rasam/ curd/ avarakkai poriyal/ chapati/ DB rice/ dal arahar kari/ wheat dosa
Tuesday	Cut - fruits/millet bisibelle-bath/coconut chutney	Soup minestoni soup or juice (buttermilk)	Kadalai curry/ paneer kofta curry/ bindi do pyaza/ chapati/ double boiled rice/ plain curd	Mix sprout salad/ southekeyi sambar/ pineapple rasam/ curd/ koridai peanut/ carrot dhaniya chapati/ DB rice/ dal masoor palak/ wheat dosa
Wednesday	Cut - fruits/Poha/Roasted Chana chutney	Soup broccoli / juice (papaya)	Lobiya masaldar/ ridge gourd korma/ beans/ chapati/ double boiled rice/ plain curd	Greek vegetable salad/ knol khol sambar/ lemon rasam/ curd/ dadamba poriyal/ chapati/ DB rice/ jeera moong dal tadka/ wheat dosa
Thursday	Cut fruits/millet dosa/peanut chutney	Soup roasted lentil soup / juice (watermelon)	Dal tiranga/ paneer spring masala/ keerai masiyal/ chapati/ double boiled rice/ plain curd	Cucumber and peanut salad/ drumstick sambar/ dal rasam/ curd/ bitter gourd masala/ jeera haldi chapati/ DB rice/ dal sabbaki/ wheat dosa
Friday	Cut fruits/vegetable rava khichidi/coconut chutney	Soup cream of tomato or juice (fresh lime water)	Dal palak/ vendakai pulusu/ mix veg avial/ chapati/ double boiled rice/ plain curd	Kachumber salad/ vendakai sambar/ ginger rasam/ curd/ chow chow poriyal/ kalonji pudina chapati/ DB rice/ lasooni dal tadka/ wheat dosa
Saturday	cut fruits/semiya upma/coconut chutney	Soup grilled zucchini & leeks or juice (muskmelon)	Methi dal/ subnam curry/ long beans thoran/ DB rice/ plain curd	Quionova bell pepper salad/ mix veg sambar/ tomato rasam/ curd/ cabbage & beans thoran/ coriander and jeera chapati/ DB rice/ dal sabbaki/ wheat dosa

FOOD AND BEVERAGE DEPARTMENT

The hospital catering service (F & B) is an essential part of the hospital operation and patient care. It's specific diet requirements and preferences.



QUALITY CONTROL

Quality control of food:

- It is very important to maintain the quality of food. Breakfast, lunch and dinner are tasted by the dietitian and the rating is recorded daily.
- Well balanced nutritious meal menu is prepared for the whole week; it is altered every 3-6 months to provide a variety to the patients.
- Regular health checkups are conducted for the kitchen staff.
- Staff are regularly trained in safe food handling.
- The sample of the patient food is retained 24 hours.

Quality control of area.

- Schedule of kitchen cleaning is made for kitchen cleaning, and the record is maintained for the same.
- Maintenance and calibration of equipment is done by the maintenance department and biomedical department periodically.
- Monitoring of deep freezers and refrigerators is done daily, and it is recorded.
- Wet and dry waste is segregated, which is emptied twice daily.
- Soiled trays and plates are put in separate bins and transferred to the washing area.
- An efficient chimney system has been placed for ventilation of hot smoke from the stoves.

FUNCTIONING OF THE KITCHEN:

- **Purchasing:** Perishables are purchased and prepared within an appropriate time frame to prevent spoilage. Items are rotated on a FIFO (First In, First Out) basis.
- **Receiving:** When products are received. Each item is checked for the correct amount. weight. price. size and damage or adulteration. Commodities are dated and placed in a rotating system to ensure FIFO.
- **Dry Storage:** All products are stored at least 6 inches off the floor and 18 inches from the ceiling. FIFO is as follows.
- **Preparation:** To maintain the quality of food served, raw materials are sourced from a known vendor. Good and hygiene methods of preparation are adhered to. Standardized cooking methods like pressure cooking, shallow fat frying in good quality utensils is followed by the cook.
- **Health:** Regular health checkup and training in food handling of the kitchen staff.
- **Food Production and Service:** It encompasses the preparation of different types of routine hospital diets, which include normal, renal, diabetic, semi-solid, soft bland and liquid diet. The setting and service of the diet varies in general, private, semi-

private and special ward. Planned meal cards are sent to the kitchen; tray setting is done and distributed to the respective wards.

Service timings	Meal
6:00am - 6:30am	Tea/coffee
7:00am - 8:00am	Breakfast
10:00am - 10:30am	Midmorning
1 :00pm -1 :30pm	Lunch
4:00pm - 4:30pm	Tea with snacks
7:30pm - 8:30pm	Dinner
9:30pm	Bedtime

- **Stores:** The store is situated in a separate room. food is stored on racks. FIFO is followed, perishable foods are refrigerated, and records are maintained.
- **Hospital Canteen:** It is on the 4th floor that caters for the staff and all the visitors. The employees can avail themselves of the services provided in the canteen at a subsidized rate.
- **Hospital Coffee Shop:** It is located on the ground floor where coffee/ tea. sandwiches. rolls, cakes and other snacks are available for all
- **Cleaning:** This is a continuous ongoing process. cleaning and sanitation of all the food surfaces and equipment and utensils are done with water. detergent and appropriate disinfectants are used. Daily and weekly cleaning schedules are followed by the staff.

DIFFERENT CHEMICALS USED FOR SPECIFIC AREA:

- Task, RT- walls, tiles
- Suma rinse/ Suma nova- dish wash
- Suma grill-hoods, exhaust, griller/fryer, bain-marie, refrigerator.
- Suma tab-knives or chopping board
- Hypo mag-utensils

TOP 3 SAFETY DEFENCE:

1. Food workers with good personnel hygiene.
2. Food cooked or held at the correct temperature.
3. Prevention of cross contamination.
 - Proper temperature is required for the safety of potentially hazardous food.
 - A thermometer must be used to make sure the cooked or cooled or stored food is at the correct temperature. Most bacteria do not grow in hot or cold

temperatures. To keep the food safe, cold food must be kept at 40°F/ 4°C or cooler. Hot foods must be kept at 150°F/ 65°C or above.

- The range of temperature between 41°F/ 5°C and 140°F/ 60°C is called the danger zone, as in this range of temperature, bacteria grow fast in the food and can cause sickness when consumed.
- **Cross contamination:** It is the transfer of harmful bacteria from one food item to another food item by means of non-food items like utensils, cutting boards, mixing bowls, counter tops, human hands etc. The equipment must be cleaned with hot water and detergents regularly. Separate cutting boards and knives should be used for different fruits and vegetables.
- **Waste disposal:**

Food waste from the kitchen and food trays from wards is collected in a trash bin lined with black liner

Black lined with collected food is removed from the trash bin, tied when they are 3/4 full and stored in the designed area

Garbage bags are transported in the specific trolley to the garbage storage area and then collected by the municipal authority.

- **Pest control:** Daily spraying of kitchen cupboards, was area and kitchen corners at night is done. Hospital pest control service. rat traps to eliminate rodents in the storage area are done on a weekly basis.
- **Records maintained by the kitchen supervisor:**
 1. **Nurse requisition sheet-** It is sent by the nursing staff for the respective wards. It contains information on new admissions, diagnosis and consultant name and diet order. It also has a list of patients on NO (Nil per oral) and changes on diet and transfer information. This sheet is checked by the dietician in the morning shift and respective changes are made before going on rounds.
 2. **Night logbook-** This book is meant to write information regarding changes of diet, new admissions and bed transfers of the inpatients and information regarding NPO is written.

MEDICAL WASTE MANAGEMENT

There are two types of waste generated in hospitals:

- Biomedical waste management by Medicare Environment Management Pt Ltd.
- General waste managed by BBMP
- Medicare Environmental Management Private Limited, the leader in waste and environment management, is the pioneer in emphasizing the need to set up scientific common bio medical waste treatment and disposal facilities.

BIO-MEDICAL WASTE SEGREGATION:

Biomedical waste must be segregated at the point of generation using color-coded containers as per regulatory guidelines.

- Black Bag – Cytotoxic Waste

Used for disposal of: Cytotoxic drugs and related waste expired medicines

Radioactive or hazardous pharmaceutical substances

This category includes waste that may be toxic, carcinogenic, or hazardous to health.

FOOD AND DRUG INTERACTION

Drugs/medicines given to patients in the hospital need to be judiciously administered. Certain drugs interact with certain foods and make the drug ineffective. Also, certain nutrients in the food consumed by the patients interact with the drug and are not absorbed by the body resulting in the patients becoming nutritionally deficient.

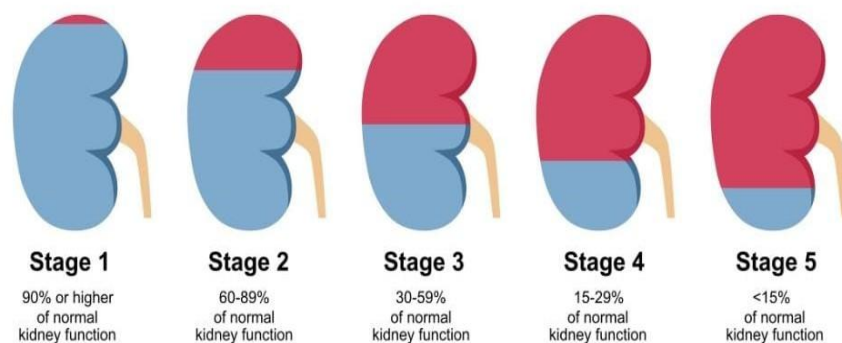
Category	Drug	Interaction
Arthritis, pain	• Analgesic • Non-steroidal Anti-inflammatory drugs (NSAIDs) • Corticosteroids	• Caffeine – Increases absorption. • Garlic, ginger, chestnut. • Limit caffeine, grapefruit and other citrus foods.
Allergies	Antihistamines	Apple, orange and grapefruit juice
Asthma	Bronchodilator	Avoid caffeine
Cardiovascular diseases	• Diuretics • Cholesterol lowering • Beta blockers • ACE inhibitors • Anti-coagulants	• Banana, orange and Green leafy vegetables. • Grapefruit, citrus fruits. • Orange juice • Banana, orange, Green leafy vegetables. • Foods high in fiber. • Vitamin K foods
Infection	• Anti-bacterial • Quinolones • Macrolides • Sulfonamides • Tetracycline's • Anti-fungal	• Vitamins • Caffeine • Citrus fruits and carbonated drinks • Dairy products • Grapefruits and citrus fruits
Seizures	Anti-convulsant / anti-epileptic therapy	Grapefruit, citrus fruit, star fruit or pomegranate juice

CASE STUDY – 1

CHRONIC KIDNEY DISEASE

INTRODUCTION: Chronic Kidney Disease (CKD) – Overview
Chronic Kidney Disease (CKD) is a progressive, irreversible loss of kidney function lasting for more than 3 months. It is characterized by a gradual decline in glomerular filtration rate (GFR) and/or evidence of kidney damage such as albuminuria. CKD is classified into five stages based on GFR, with Stage 5 being the most severe form, also known as End-Stage Renal Disease (ESRD), where kidney function is less than 15 mL/min/1.73 m² and renal replacement therapy becomes necessary.

Stages of Chronic Kidney Disease



The kidneys contain functional units called nephrons, responsible for filtration, electrolyte balance, and waste removal.

In CKD:

- Progressive nephron loss occurs.
- Remaining nephrons undergo hyperfiltration.
- Glomerulosclerosis develops.
- Toxins (urea, creatinine) accumulate.
- Fluid and electrolyte imbalance occurs.
- Hormonal functions (erythropoietin, vitamin D activation) reduce.
- This ultimately leads to uremia and systemic complications.

Etiology (Causes):

- Chronic Kidney Disease Stage 5 develops due to long-standing damage to the kidneys over several years.
- Uncontrolled diabetes mellitus is the most common cause and leads to progressive damage to the glomeruli.
- Chronic hypertension causes thickening and narrowing of renal blood vessels, reducing kidney function.
- Chronic glomerulonephritis results in inflammation and scarring of the glomeruli.
- Polycystic kidney disease causes gradual destruction of normal kidney tissue by multiple cysts.
- Recurrent kidney infections and obstructive uropathy contribute to progressive renal damage.
- Autoimmune disorders such as systemic lupus erythematosus can damage kidney tissues.
- Long-term use of nephrotoxic drugs such as NSAIDs may accelerate kidney failure.

Risk Factors:

- Individuals with long-standing diabetes are at high risk of developing CKD.
- Poorly controlled hypertension significantly increases the risk of kidney damage.
- A positive family history of kidney disease predisposes individuals to CKD.
- Obesity increases the workload on the kidneys and contributes to disease progression.
- Smoking accelerates vascular damage and worsens kidney function.
- Cardiovascular disease increases the likelihood of renal complications.
- Recurrent urinary tract infections increase the risk of chronic kidney damage.
- Advanced age (above 60 years) is associated with declining renal function.

Clinical Manifestations (Stage 5):

- Patients experience severe fatigue and weakness due to anemia and toxin accumulation.
- Edema develops in the legs, face, and abdomen due to fluid retention.
- Urine output decreases significantly and may progress to anuria.
- Patients commonly experience nausea, vomiting, and loss of appetite.
- A metallic taste in the mouth and bad breath may occur due to uremia.
- Shortness of breath develops due to fluid overload in the lungs.
- Persistent itching (pruritus) occurs due to toxin buildup.
- Muscle cramps are common due to electrolyte imbalance.
- Confusion and altered mental status may occur in advanced uremia.

Diagnosis:

- CKD Stage 5 is diagnosed when GFR falls below 15 mL/min/1.73 m² for more than three months.
- Serum creatinine and blood urea levels are significantly elevated.
- Hyperkalemia and hyperphosphatemia are commonly observed.
- Metabolic acidosis may be present.
- Hemoglobin levels are reduced due to decreased erythropoietin production.
- Urine examination shows persistent proteinuria.
- Ultrasound imaging reveals small, shrunken kidneys with loss of normal architecture.

Management:

- Renal replacement therapy such as hemodialysis or peritoneal dialysis is required for survival.
- Kidney transplantation is considered the definitive treatment option.
- Blood pressure is controlled by using antihypertensive medications such as ACE inhibitors or ARBs.
- Anemia is treated with erythropoietin-stimulating agents and iron supplementation.
- Phosphate binders are prescribed to manage hyperphosphatemia.
- Vitamin D supplementation is given to prevent bone disorders.
- Sodium bicarbonate is administered to correct metabolic acidosis.
- Fluid and electrolyte levels are regularly monitored.

Treatment of Chronic Kidney Disease (CKD):

Treatment of CKD focuses on slowing disease progression, managing complications, and replacing kidney function in advanced stages.

1) Treatment to Slow Disease Progression (Early to Moderate CKD):

- Blood pressure is controlled and maintained below 130/80 mmHg to reduce further kidney damage.
- ACE inhibitors or ARBs are prescribed to reduce proteinuria and protect kidney function.
- Blood glucose levels are strictly controlled in diabetic patients to prevent diabetic nephropathy progression.
- Nephrotoxic drugs such as NSAIDs are avoided to prevent additional kidney injury.
- Regular monitoring of GFR, serum creatinine, and urine protein is done to assess progression.

2. Management of Complications:

a) Anemia Management

Erythropoietin-stimulating agents are administered to stimulate red blood cell production.

Iron supplementation is given if iron deficiency is present.

b) Electrolyte Imbalance

Hyperkalemia is treated with dietary potassium restriction and potassium-lowering medications.

Metabolic acidosis is corrected using sodium bicarbonate therapy.

c) Bone and Mineral Disorders

Phosphate binders are prescribed to control hyperphosphatemia.

Active vitamin D supplements are given to manage hypocalcemia and prevent renal osteodystrophy.

d) Fluid Overload

Diuretics are prescribed if urine output is present.

Sodium and fluid intake are restricted to prevent edema and hypertension.

3. Renal Replacement Therapy (CKD Stage 5):

When GFR falls below 15 mL/min/1.73 m², renal replacement therapy becomes necessary.

- Hemodialysis removes waste products and excess fluid using a dialysis machine.
- Peritoneal dialysis uses the peritoneal membrane as a filter to remove toxins.
- Kidney transplantation is considered the definitive treatment and improves long-term survival and quality of life.

Medical Nutrition Therapy (MNT):

- Energy intake of 30–35 kcal/kg/day is recommended to prevent protein breakdown.
- Protein intake is restricted to 0.6–0.8 g/kg/day in pre-dialysis patients.
- Protein intake is increased to 1.0–1.2 g/kg/day in patients on dialysis.
- Sodium intake is restricted to less than 2 grams per day.
- Potassium intake is limited according to serum potassium levels.
- Phosphorus intake is restricted to 800–1000 mg per day.
- Fluid intake is calculated as urine output plus 500 mL in dialysis patients.
- High biological value protein sources are preferred in controlled amounts.

Prevention:

Medical Prevention:

- Strict control of blood glucose levels prevents diabetic nephropathy progression.
- Maintaining blood pressure below 130/80 mmHg reduces kidney damage.
- Regular screening of high-risk individuals helps in early detection.
- Avoidance of nephrotoxic drugs prevents further renal injury.

Dietary Prevention:

- Moderate protein intake slows the progression of kidney damage.
- Reducing sodium intake helps control blood pressure and edema.
- Limiting processed foods reduces phosphorus and potassium overload.
- Adequate calorie intake prevents muscle wasting.

Lifestyle Prevention:

- Maintaining healthy body weight reduces kidney workload.
- Regular physical activity improves overall cardiovascular health.
- Smoking cessation slows the progression of kidney disease.
- Limiting alcohol intake protects renal function.
- Regular medical follow-up improves disease monitoring and management.

CASE PROFILE:

NAME	MR. TABREZ AHMED
AGE	50 yrs
GENDER	MALE
FOOD HABITS	NON-VEGETARIAN
OCCUPATION	BUSINESS
MEDICAL CONDITION	CHRONIC KIDNEY DISEASE
PRESENT COMPLAINTS	BREATHLESSNESS, FEVER, HYPERTENTION
CO-MORBIDITIES	HYPERTENTION
FAMILY HISTORY	NIL
PAST MEDICAL HISTORY	NIL
ALLERGIES	NIL

ADMISSION DETAILS:

09 Feb 2026

DISCHARGE DETAILS:

11 Feb 2026

ANTHROPOMETRIC MEASUREMENTS:

Height in cm	152.4 cm
Weight in kg	54.6 kg
BMI (kg/m)	23.63 kg/m

Interpretation

According to Asian classification of BMI, the patient is overweight.

BIOCHEMICAL PARAMETERS:

PARAMETERS	PATIENTS VALUE	REFERENCE RANGE	INTERPRETATION
Hemoglobin	7.6 g/dL	13-16 g/dL	The patient has low hemoglobin
Packed cell volume	24.9%	36-48%	The patient has low PCV
Total volume	3530 cumm	4000-10000 cumm	The patient has low WBC
RBC count	4.46 cumm	4.5-5.5 cumm	The patient's RBC is slightly decreased
Mean corpuscular	55.8 fL	78.86 fL	The patient has low MCV
Mean corpuscular hemoglobin	17.0 pg	28-34 pg	The patient has low MCH
Mean corpuscular hemoglobin concentration	30.5 g/dL	33-36 g/dL	The patient has low MCHC
Red cell distribution width	19.2%	11.6- 14%	The patient's RDW-CV is increased
Lymphocytes count	18.1%	25-45%	The patient has low lymphocyte count
Basophils	1.1%	0-1%	The patient has high basophil count

MEDICATIONS:

DRUG	DOSAGE	FREQUENCY	DURATION
Inj. Erythropoietin	10000 units	S/C (3 times / week)	To continue
Tab.Ceftum	250mg	1-0-1	For 5 days
Cap. Fluvir	75mg	1-0-0	For 3 days
Tab. Thyronorm	150mcg	1-0-0	To continue (in empty stomach)
Tab. Shecal	500mg	1-0-0	To continue (alternate days)
Tab. Lasix	40mg	2-2-0 (8am-4pm)	To continue
Tab. Ecosprin	75mg	0-1-0	To continue
Tab. Calcitriol	0.25mg	2-0-0	To continue
Tab. Rivaroxaban	2.5mg	1-0-0	To continue
Tab. Nodon	2.5mg	1-0-0	To continue

DIETARY ASSESSMENT:**24-hour dietary recall**

TIME	MEAL	MENU	PORTION SIZE	AMOUNT(g)
11 – 11:30 AM	Breakfast	Idly sambar	3 – 4 pieces	120g
3:00 PM	Lunch	Rice, dal, chicken	1 bowl 1 bowl 1 bowl	60g 60g 100g
5 – 6: PM	Evening snacks	Tea and biscuit	1 cup	150ml
10 – 10:30 PM	Dinner	Roti, Chicken	1 roti 1 bowl	30g 100g

HOSPITAL DIET:

TIME	MEAL	MENU	PORTION SIZE	AMOUNT(g)
11 – 11:30 AM	Breakfast	Idly sambar	3 – 4 pieces	120g
3:00 PM	Lunch	Rice, dal, palya	1 bowl 1 bowl 1 bowl	60g 60g 100g
5 – 6: PM	Evening snacks	Tea and biscuit	1 cup	150ml
10 – 10:30 PM	Dinner	Roti, palya	1 roti 1 bowl	30g 100g

Nutritional screening:

Height	152.4cm
Weight	54.6kg
BMI	$54.6/(1.52)^2 = 54.6/2.31 = 23.63\text{kg/m}^2$
Interpretation	Over-weight
Ideal body weight Hamwi's method	106lb for first 5 feet+6 for each additional inch 106lb $106\text{lb} \div 2.24 = 47.32\text{kg}$ (1kg=2.24lb)
Broca's Index	Height in cm- 152.4-100 = 52.4kg
Harris benedict equation	$\text{REE} = 66 + (13.7 \times \text{weight in kg}) + (5 \times \text{height in cm}) - (6.8 \times \text{age})$ $= 66 + (13.7 \times 54.6) + (5 \times 152.4) - (6.8 \times 50)$ $= 66 + (748.02) + (762) - (-340)$ $1576.02 - 340 = \mathbf{1236.02 \text{ kcal}}$ $\mathbf{TEE = REE \times PAL = 1236.02 \times 1.4 = 1730.42 \text{ kcal}}$

NUTRIENT RECOMMENDATION:

Nutrients	Guidelines	Modified Values
Energy	30 - 35 kcal/kg/body weight	1730.42 (33kcal/kg/body weight)
Carbohydrate	-	74% = 320.12g
Proteins	10-15%	15% = 64.89g
Fats	20-25%	20% = 38.45g
Calcium	800-1200mg	800mg
Iron	8-10mg	9mg
Potassium	1500-3400mg	1560mg
Sodium	1200-3000mg	1200mg

Diet Plan:

MEAL	MENU	INGREDIENTS	AMOUNT (g/ml)	ENERGY (Kcal)	PROTEIN (g)	CARB (g)	FAT (g)
EARLY MORNING	FRUIT and nut	green apple	100g	51.19	0.46	10.65	0.5
		almond	10G	60.98	1.84	3.57	5.84
BREAKFAST	ragi dosa	ragi	30g	96.31	2.41	20.04	0.57
	boiled egg	parboiled rice	60g	214.01	4.76	46.94	0.31
		egg	100g	52	12.37		0.26
MID MORNING	salad	cucumber	50g	9.8	0.35	1.74	0.08
		paneer	50g	129.06	9.43	6.2	7.39
LUNCH	chicken biryani	rice	60g	214.01	4.76	46.94	0.31
	raita	tomato	50g	9.8	0.44	1.34	0.22
		onion	50g	24.04	0.74	2.78	0.12
		chicken	100g	383.97	19.44		12.64
		curd	50g	36.48	1.63	2.47	2.24
MID EVENING	gingelly seed chikki	gingelly seeds	10g	52	2.16	1.08	4.3
		jaggery	5g	17.7	0.09	4.24	0
DINNER	roti	wheat	30g	105.64	0.04	22.28	0.22
	tinda	tinda	100g	13.87	1.02	1.9	0.17
		onion	50g	24.04	0.74	2.78	0.12
		tomato	50g	9.8	0.44	1.34	0.22
	total oil	safflower oil	20ml	180			6.17
		TOTAL		1684.7	63.12	176.29	41.68

MEAL	MENU	INGREDIENTS	AMOUNT (g/ml)	CALCIUM (mg)	IRON (mg)	POTASSIUM (mg)	SODIUM (mg)
EARLY MORNING	FRUIT and nut	green apple	100g	6.53	0.2	94.55	1.47
		almond	10G	22.8	0.45	69.9	0.15
BREAKFAST	ragi dosa	ragi	30g	109.2	1.38	132.9	1.42
	boiled egg	parboiled rice	60g	4.49	0.39	85.2	1.89
		egg	100g	8.07	0.15	147	144
MID MORNING	salad	cucumber	50g	8.19	0.23	91.5	3.16
		paneer	50g	238	0.45	31.76	9.02
LUNCH	chicken biryani	rice	60g	4.49	0.39	64.8	1.4
	raita	tomato	50g	5.08	0.14	102	1.4
		onion	50g	10.5	0.2	85.5	2.75
		chicken	100g	20.52	1.27	283	65.07
		curd	50g	59	0.07	57.5	12.73
MID EVENING	gingelly seed chikki	gingelly seeds	10g	128.3	1.5	46	1.54
		jaggery	5g	5.35	0.23	24.4	1.26
DINNER	roti	wheat	30g	6.12	5.3	93.3	0.16
	tinda	tinda	100g	19.68	0.41	56.18	20.61
		onion	50g	10.5	0.2	85.5	2.75
		tomato	50g	5.08	0.14	102	1.4
	total oil	safflower oil	20ml				
		TOTAL		671.9	13.1	1652.99	272.18

Modified table:

Nutrient	Modified value	Amount computed through diet plan	Difference
Energy	1730.4kcal	1684.7kcal	-45.72
Carbohydrate	320.12g	176.29g	-143.83
Protein	64.89g	63.12g	- 1.77
Fat	39.45g	41.68g	+ 2.23
Calcium	800mg	671.9mg	- 128.1
Iron	9mg	13.1mg	+ 4.9
Potassium	1560mg	1652.99mg	+ 92.99
Sodium	1200mg	272.18mg	- 927.83

Justification:

This diet plan has been carefully designed for a 50-year-old man with chronic kidney disease (CKD) complicated by hypertension, breathing difficulty, and hypothyroidism, taking into account his age, metabolic changes, and increased cardiovascular risk. At 50 years, renal function decline combined with comorbidities increases the risk of protein-energy wasting, fluid overload, electrolyte imbalance, and cardiac complications. Therefore, energy intake is kept adequate to prevent muscle breakdown and fatigue, particularly important because hypothyroidism lowers basal metabolic rate and can contribute to lethargy. Protein is provided in controlled amounts, mainly from high biological value sources such as egg and chicken in measured portions, to support tissue maintenance while minimizing nitrogenous waste accumulation and preventing further renal burden.

Sodium intake is restricted to manage hypertension and reduce fluid retention, which is crucial in this patient as excess fluid can worsen breathlessness and increase the risk of pulmonary congestion. Potassium-containing foods are included in regulated quantities to prevent hyperkalemia, a serious complication in CKD. Fat intake is moderated to reduce cardiovascular risk, which is higher in a 50-year-old male with hypertension and renal disease. Carbohydrates are provided in sufficient amounts to supply energy and prevent protein catabolism, thereby protecting lean body mass. Calcium and iron sources are included in appropriate amounts to address the risk of CKD-related anemia and bone mineral imbalance.

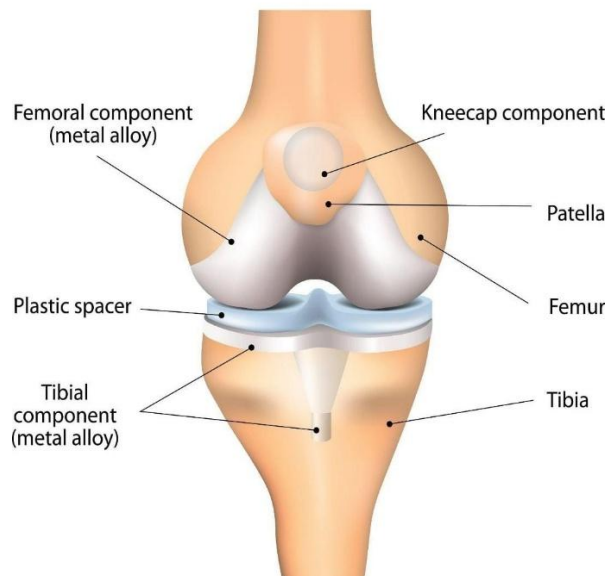
Overall, this diet aims to slow progression of kidney disease, maintain electrolyte and fluid balance, control blood pressure, reduce respiratory distress related to fluid overload, and support metabolic stability in a middle-aged man with multiple chronic conditions, ensuring both renal protection and overall systemic health.

CASE STUDY – 2

TOTAL KNEE REPLACEMENT (TKR)

INTRODUCTION: Total Knee Replacement (TKR), also known as total knee arthroplasty, is a surgical procedure in which the damaged articular surfaces of the knee joint are removed and replaced with artificial prosthetic components. It is commonly performed to relieve severe pain, improve joint function, and enhance quality of life in patients with advanced knee joint disease that does not respond to conservative treatment.

TOTAL KNEE IMPLANT



Anatomy of the knee joint:

The knee joint is a hinge joint formed by the femur, tibia, and patella. It contains articular cartilage, synovial fluid, ligaments, and menisci that allow smooth movement. In degenerative conditions such as osteoarthritis, the cartilage wears down, leading to pain and stiffness. In TKR surgery, the damaged surfaces are replaced with metal and polyethylene components.

Etiology (causes):

- Total Knee Replacement is mainly performed due to severe degeneration of the knee joint.
- Osteoarthritis is the most common cause and results in gradual wear and tear of the articular cartilage.
- Rheumatoid arthritis causes chronic inflammation that leads to destruction of joint structures.
- Post-traumatic arthritis develops after fractures, ligament injuries, or meniscal damage to the knee.
- Avascular necrosis can lead to collapse of bone surfaces within the knee joint.
- Congenital deformities and long-standing joint instability may contribute to progressive joint damage.

Risk Factors:

- Increasing age, especially above 50 years, significantly increases the risk of knee joint degeneration.
- Female gender has a higher prevalence of osteoarthritis.
- Obesity increases mechanical stress on the knee joint and accelerates cartilage breakdown.
- Previous knee injuries or surgeries increase the likelihood of developing secondary arthritis.
- A family history of osteoarthritis predisposes individuals to joint degeneration.
- Sedentary lifestyle and occupations involving repetitive knee strain increase the risk.
- Metabolic disorders such as diabetes may contribute to joint deterioration.

Clinical Manifestations:

- Patients experience persistent knee pain that worsens with walking, climbing stairs, or standing.
- Joint stiffness limits the range of motion and affects daily activities.
- Swelling and tenderness around the knee joint are commonly observed.
- Deformities such as varus (bow-legged) or valgus (knock-knee) alignment may develop.
- Crepitus or a grinding sensation may be felt during movement.
- In advanced cases, pain may persist even at rest or during the night.

Diagnosis:

- Diagnosis is based on detailed clinical examination and patient history.
- Physical examination reveals joint tenderness, swelling, reduced mobility, and deformity.
- X-ray imaging shows joint space narrowing, osteophyte formation, and bone changes.
- MRI may be performed to assess soft tissue involvement when necessary.
- Blood tests are conducted to rule out inflammatory or infectious causes.
- TKR is recommended when conservative treatment fails and joint damage is severe.

Management:

- Initial management includes analgesics and non-steroidal anti-inflammatory drugs for pain relief.
- Physiotherapy is recommended to strengthen muscles and improve joint function.
- Weight reduction is advised to decrease stress on the knee joint.
- Intra-articular corticosteroid injections may provide temporary relief.
- Total Knee Replacement surgery is performed when conservative measures are ineffective.
- During surgery, damaged cartilage and bone are replaced with prosthetic components.
- Postoperative rehabilitation and physiotherapy are essential for recovery.
- Pain control and infection prevention are important aspects of postoperative care.

Medical Nutrition Therapy (MNT):

- Adequate energy intake is necessary to promote wound healing and recovery after surgery.
- Sufficient protein intake supports tissue repair and prevents muscle loss.
- A balanced diet rich in calcium and vitamin D supports bone health.
- Vitamin C and zinc intake promotes wound healing and immune function.
- Maintaining a healthy body weight reduces stress on the prosthetic joint.
- Gradual weight reduction is recommended in obese patients before surgery.
- Adequate fluid and fiber intake help prevent postoperative constipation.

Prevention:

1) Medical Prevention:

- Early diagnosis and proper management of osteoarthritis help slow joint degeneration.
- Appropriate use of analgesics and anti-inflammatory medications reduces chronic inflammation and pain.
- Disease-modifying anti-rheumatic drugs (DMARDs) help prevent joint destruction in patients with rheumatoid arthritis.
- Intra-articular injections such as corticosteroids or hyaluronic acid may reduce inflammation and improve joint lubrication.
- Regular physiotherapy prescribed by a healthcare professional strengthens supporting muscles and stabilizes the joint.
- Timely treatment of knee injuries prevents long-term cartilage damage.
- Regular follow-up with an orthopedic specialist helps monitor disease progression and modify treatment accordingly.

2) Dietary Prevention:

- Maintaining a healthy body weight through a balanced diet reduces mechanical stress on the knee joint.
- A diet rich in calcium and vitamin D supports bone strength and joint health.
- Adequate protein intake helps maintain muscle mass, which supports and protects the knee joint.
- Anti-inflammatory foods such as fruits, vegetables, nuts, and whole grains may help reduce joint inflammation.
- Limiting processed foods and excessive sugar intake may help reduce systemic inflammation.
- Adequate hydration supports joint lubrication and overall musculoskeletal health.

3) Lifestyle Prevention:

- Regular low-impact exercises such as walking, swimming, and cycling improve joint mobility and muscle strength.
- Avoiding high-impact activities such as jumping or running on hard surfaces protects knee cartilage.
- Maintaining proper posture and ergonomic body mechanics reduces strain on the knee joint.
- Using supportive footwear reduces abnormal pressure on the knees.

- Avoiding prolonged standing or repetitive knee bending reduces wear and tear.
- Smoking cessation improves blood circulation and delays joint degeneration.
- Weight management through consistent physical activity and healthy habits significantly reduces the risk of severe osteoarthritis.

CASE PROFILE:

Name	K.M. Mujeeb Ur Rahman
Age	70 years
Gender	Male
Food Habits	Non-vegetarian
Occupation	Nil
Present Complaints	Hypertension, Prediabetes, Dyslipidemia, Gastric erosions, Duodenal polyp, Large Hiatus hernia, Iron deficiency anemia, Mallampati Class III
Co-morbidities	CABG, Hypertension (HTN), Prediabetic
Family History	Nil
Past Medical History	Nil

ADMISSION DETAILS:

09 Feb 2026

DISCHARGE DETAILS:

16 Feb 2026

ANTHROPOMETRIC MEASUREMENTS:

Height in cm	160.02 cm
Weight in kg	84 kg
BMI (kg/m²)	33.59 kg/m ²

Interpretation: According to Asian classification of BMI, the patient is obese.

BIOCHEMICAL PARAMETERS:

PARAMETERS	PATIENTS VALUE	REFERANCE RANGE	INTERPRETATION
Hemoglobin	11.3g/dL	13-17g/dL	The patient has low hemoglobin
Packed cell volume	34.4%	36-48%	The patient has low PCV
Total count	14360 cumm	4000-11000 cumm	The patient has high total count
RBC count	3.72 cumm	4.5-5.5 cumm	The patient's RBC is slightly decreased
Mean corpuscular	92.5 fL	78-86 fL	The patient has high MCV
Mean corpuscular hemoglobin concentration	32.8 g/dL	33-36 g/dL	The patient has low MCHC
Neutrophils	90.1%	45-75%	The patient has high neutrophil
Lymphocytes count	6.1%	25-45%	The patient has low lymphocyte count
Urea	45mg per dL	19-43mg per dL	The patient has high Urea
Sodium serum	136mmol per L	137-145mmol per L	The patient has low sodium serum

CLINICAL ASSESSMENT:

Parameters	Patient Value	Reference value	Interpretation
Blood Pressure	140/90mmHg	120/80mmHg	Stage 2 Hypertension (Elevated systolic BP)
Pulse	90/min	60-100/min	Normal
CVS	S1, S2(+)	-	Normal heart sounds
RS	Clear	-	Normal breath sounds
SpO2	94%	95-100%	Slightly low

FINAL DIAGNOSIS:

Osteoarthritis of Both Knees.

Newly Detected Diabetes Mellitus

HISTORY OF PRESENT ILLNESS:

Mr. K M. Mujeeb Ur Rahman, 70 years old male, presented with the complaints of pain in both knees which was insidious in onset gradually progressive non radiating non referring with no diurnal variation and aggravated on activities and relieved on taking medications. Patient also complaints of unable to squat, stair climbs, sitting cross legged. Hence, admitted for further evaluation and management.

PAST HISTORY:

Known case of Hypertension, Ischaemic Heart Disease - (S/P PTCA) - on regular medications

Septoplasty-15 years back
Esophageal Banding
History of Bronchial Asthma, OSA, Orthopnea.

LOCAL EXAMINATION:

BILATERAL KNEE

Medial joint line tenderness (+)
Varus deformity (+)
Crepitus (+)
ROM 0-100 degrees
No DNVD

INVESTIGATION: Enclosed

PRE OP X-RAY:

Decreased joint space, sclerosis of joint margins, multiple osteophytes with varus deformity in both knees suggestive of osteoarthritis.

POST OP X-RAY:

Total Knee Replacement implants in situ in both knees with satisfactory alignment.

PROCEDURE:

Total knee replacement on right side done on 10.02.2026
Total knee replacement on left side done on 13.02.2026

ANAESTHESIA NOTES:

Surgery was performed under SA for right Knee and Epidural left Knee. Intra operative period and post-operative period uneventful.

PROCEDURES 1:

Total knee replacement on right side done on 10.02.2026 using DePuy's PFC Sigma PS implants.

RIGHT SIDE (IMPLANT DETAIL):

Femoral posterior stabilized cemented - Size 3 Right
Modular Tibial Tray - Size 3 cemented

Tibial Insert fixed bearing stabilized - 3, 10 mm GVF
Gentamicin bone cement: 40gm

PROCEDURE 2:

Total knee replacement on left side done on 13.02.2026 using DePuy's PFC Sigma PS implants.

LEFT SIDE (IMPLANT DETAIL):

Femoral posterior stabilized cemented - Size 3 Left

Modular Tibial Tray - Size 3 cemented

Tibial Insert fixed bearing stabilized - 3, 12.5 mm GVF

Gentamicin bone cement: 40gm

Patient withstood the procedure well. Patient was transfused with 1 unit PRBC. In view of elevated sugar levels physician opinion was sought and found to have Newly detected Diabetes Mellitus and advise followed. Gastroenterologist opinion was sought and advise followed. Patient was ambulated full weight bearing with walker and knee brace. Started on ROM knee and quadriceps exercises from 1st post-operative day. Wound is clean and healthy at the time of discharge.

MEDICATIONS:

DRUG	DOSAGE	FREQUENCY	DURATION
Tab. Cetil	500 mg	1-0-1	for 5 days
Tab. Ultracet	—	1-0-1	for 5 days & there after whenever pain occurs - 20 tabs
Tab. Pan	40 mg	1-0-0	for 5 days (before food)
Cap. Becosules	—	0-1-0	for 2 weeks
Tab. Ecosprin	150 mg	0-1-0	for 4 weeks
Tab. Glimy	1mg	1-0-1	to continue (after food)
Cap. Nexpro RD	—	1-0-1	for 5 days (Followed by)
—	—	1-0-0	for 15 days
Tab. Ondem MD	4mg	—	SOS

DIETARY ASSESSMENT:**24-hour dietary recall**

TIME	MEAL	MENU	PORTION SIZE	AMOUNT(g)
10:30 – 11AM	Breakfast	Roti Veg palya	1 Roti 1 Bowl palya	30g 60g
2:30 - 3PM	Lunch	Dal, Rice, Veg palya	1 bowl 1 bowl 1 bowl	60g 60g 100g
5:00 PM	Evening snacks	Tea and biscuit	1 cup	150ml
9:30PM	Dinner	Roti, Veg palya	1 roti 1 bowl	30g 60g

HOSPITAL DIET:

TIME	MEAL	MENU	PORTION SIZE	AMOUNT(g)
7:30 AM	Breakfast	Idly Sambar	2-3 Pieces 1 bowl	90g 60g
12:30 PM	Lunch	Rice, Saag	1 bowl 1 bowl	30g 60g
5:00 PM	Evening snacks	Tea and biscuit	1 cup	150ml
7:30PM	Dinner	Rice, Saag	1 bowl 1 bowl	30g 60g

Nutritional screening:

Height	160.02cm
Weight	86kg
BMI	$86 / (1.60)^2 = 86 / 2.56 = 33.59\text{kg/m}^2$

Interpretation	Obese
Ideal body weight Hamwi's method	106lb for first 5 feet+6 for each additional inch 106lb+6(3) lb = 106+18 124lb ÷ 2.24 = 55.35kg (1kg=2.24lb)
Broca's Index	Height in cm 160.02-100 = 60.02kg
Harris benedict equation	REE=66+(13.7×weight in kg) + (5×height in cm) – (6.8×age) = 66+(13.7×60.02) + (5×160.02) – (6.8×70) = 66+(824.74) + (800.1) – (476) =1690.84 - 476 = 1214.84 kcal TEE=REE×PAL=1214.84×1.2=1457.80 kcal

NUTRIENT RECOMMENDATION:

Nutrients	Guidelines	Modified Values
Energy	25-30 kcal/kg/body weight	1457.80 (26kcal/kg/body weight)
Carbohydrate	-	68% = 396.57g
Proteins	15-20%	17% = 61.95g
Fats	15-20%	15% = 24.29g
Calcium	800-1200mg	1200mg
Iron	19-22mg	22mg

Diet Plan:

MEAL	MENU	INGREDIENTS	AMOUNT	ENERGY	PROTEIN	CARB	FAT	CALCIUM	IRON
EARLY MORNING	MILK	milk	100ml	72.96	3.26	4.94	4.48	118	0.15
BREAKFAST	BROKEN WHEAT UPMA	DALIA	30g	102.63	3.25	20.71	0.43	8.12	1.15
		ONION	50g	24.04	0.74	2.78	0.12	10.5	0.2
		TOMATO	50g	9.8	0.44	1.34	0.22	5.08	0.14
MID MORNING	SOUP	BOTTLE GUARD	100G	11	0.52	1.68	0.13	15.42	0.26
LUNCH	RED RICE PALAK DAL SALAD STEAMED VEG	RED RICE	30g	106.11	2.75	22.44	0.37	9.9	0.45
		SPINACH	100g	24.4	2.14	2.04	0.64	82.28	2.94
		ONION	50g	24.04	0.74	2.78	0.12	10.5	0.2
		TOMATO	50g	9.8	0.44	1.34	0.22	5.08	0.14
		TUR DAL	30g	99.33	6.51	16.56	0.46	21.51	1.17
		PANEER	50g	129.06	9.43	6.2	7.39	238	0.45
		CUCUMBER	50G	9.8	0.35	1.74	0.08	8.19	0.23
		MOONG SOROUTS	30g	96.81	7.3	15.75	0.22	13.39	2.11
		MORINGA PODS	100g	29.42	2.62	3.76	0.12	33.3	0.73
MID EVENING	FRUITS	APPLE	100g	62.44	0.29	13.11	0.64	13.68	0.26
		PEAR	100g	37.55	0.36	8.09	0.27	6.55	0.28
		ORANGE	100G	37.32	0.7	7.92	0.13	19.52	0.81
DINNER	BAJRA ROTI PALAK DAL SALAD STEAMED VEG	BAJRA	30g	104.49	3.28	18.53	1.62	8.2	1.92
		SPINACH	50g	12.2	1.07	1.02	0.32	41.14	1.47
		ONION	50g	24.04	0.74	2.78	0.12	10.5	0.2
		TOMATO	50g	9.8	0.44	1.34	0.22	5.08	0.14
		TUR DAL	30g	99.33	6.51	16.56	0.46	21.51	1.17
		PANEER	25G	64.53	4.71	3.1	3.69	119	0.22
		CUCUMBER	50G	9.8	0.35	1.74	0.08	8.19	0.23
		MOONG SOROUTS	30g	96.81	7.3	15.75	0.22	13.39	2.11
		MORINGA PODS	100g	29.42	2.62	3.76	0.12	33.3	0.73
BED TIME	SOUP	BOTTLE GUARD	100G	11	0.52	1.68	0.13	15.42	0.26
		TOTAL OIL	10ML	52.05			3.96		
		TOTAL		1399.98	69.38	199.44	26.98	894.75	20.12

Modified table:

Nutrient	Modified value	Amount Computed through diet plan	Difference
Energy	1457.8 kcal	1399.98kcal	- 57.82
Carbohydrate	396.57g	199.44g	- 197.13
Protein	61.95g	69.38g	+ 7.43
Fat	24.29g	26.98g	+ 2.69
Calcium	1200mg	894.75g	- 305.25
Iron	22mg	20.12mg	- 1.88

Justification:

This diet has been carefully planned for a 70-year-old male patient who has undergone Total Knee Replacement (TKR) surgery and is diagnosed with ischemic heart disease, prediabetes, obesity, bronchial asthma, osteopenia, and documented calcium and iron deficiency. At this age, recovery is slower, and multiple comorbidities increase the risk of complications; therefore, the diet aims to promote wound healing, support cardiovascular health, improve glycemic control, assist in weight management, reduce inflammation, and correct micronutrient deficiencies simultaneously.

Following TKR surgery, adequate protein is essential for tissue repair, collagen formation, and preservation of muscle mass, especially in elderly individuals who are prone to sarcopenia. The inclusion of moong dal, tur dal, sprouts, milk, and paneer ensures moderate but sufficient protein intake without overloading the kidneys or increasing saturated fat excessively. The total energy provided is controlled to prevent further weight gain, which is important in obesity, as excess body weight increases stress on the replaced knee joint and may delay rehabilitation.

For ischemic heart disease management, the diet restricts visible fat (limited oil use) and avoids fried, processed, and high-saturated-fat foods. Emphasis on whole grains such as broken wheat (dalia), red rice, and whole wheat roti provides complex carbohydrates and dietary fiber, which help improve lipid profile and reduce cardiovascular risk. High-fiber vegetables and fruits like apple, pear, and orange further support cholesterol reduction and improve satiety, aiding weight control.

Since the patient is prediabetic, carbohydrate intake is distributed evenly across meals to prevent sudden spikes in blood glucose levels. Low glycemic index foods and fiber-rich vegetables such as bottle gourd, cucumber, spinach, and moringa pods help in maintaining stable blood sugar levels. Small, frequent meals also reduce metabolic stress and improve insulin sensitivity.

The presence of osteopenia along with calcium deficiency makes calcium intake a major priority. Milk, paneer, and green leafy vegetables such as spinach contribute to improving calcium levels and supporting bone mineral density, which is crucial after TKR surgery for proper bone healing and long-term joint stability. Additionally, adequate vitamin and mineral intake supports overall skeletal strength and reduces fracture risk.

Iron deficiency is addressed through inclusion of iron-rich foods such as spinach, sprouts, dal, and vegetables. Improving iron intake helps prevent anemia, enhances oxygen-carrying capacity, and improves energy levels, which are necessary for physiotherapy and recovery. Adequate iron status also supports immune function and wound healing.

For bronchial asthma and esophageal concerns, the diet avoids heavy, oily, and spicy meals that may trigger reflux or breathing discomfort. Warm soups and soft-textured foods are included to make digestion easier and reduce strain. The antioxidant-rich vegetables and fruits help reduce oxidative stress and inflammation, which may support respiratory health.

Overall, this diet is balanced, heart-friendly, calorie-controlled, nutrient-dense, and tailored to correct calcium and iron deficiencies while managing chronic conditions and promoting optimal post-operative recovery in this elderly patient.